
EECS 16B Designing Information Devices and Systems II

Discussion Worksheet Discussion 12A

In this discussion, we practice computing the SVD for a "wide" matrix (more columns than rows) and for a "tall" matrix (more rows than columns). There is also an associated jupyter notebook on Datahub that will serve useful to confirm the numerical calculations (specifically for performing Gram-Schmidt).

Also, note that the techniques and insights communicated in this discussion are conveyed in [Note 13, sec. 3](#).

1. Computing the SVD: A "Tall" Matrix Example

Define the matrix

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 2 & -2 \end{bmatrix}.$$

(a) In this part, we will find the full SVD of A in steps.

(i) Compute $A^T A$ and find its eigenvalues.

(ii) Find orthonormal eigenvectors $\vec{v}_i$ (right singular vectors, columns of V).

(iii) Find singular values, $\sigma_i = \sqrt{\lambda_i}$.

(iv) Use $\vec{v}_i$ to find orthonormal $\vec{u}_i$ (for nonzero σ).

(v) Use the previous parts to write the full SVD of A .

(vi) Use the Jupyter notebook to run the code cell that calls `numpy.linalg.svd` on A . What is the result? Does it match our result above?

(b) Find the rank of A .

(c) Find a basis for the range (or column space) of A .

(d) Find a basis for the null space of A .

(e) We now want to create the SVD of $A^\top$. Rather than repeating all of the steps in the algorithm, feel free to use the jupyter notebook for this subpart (which defines a `numpy.linalg.svd` command). What are the relationships between the matrices composing A and the matrices composing $A^\top$?

2. Computing the SVD: A "Wide" Matrix Example

Define the matrix

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}.$$

- (a) In this part, we will find the full SVD of A in steps.
 - (i) Compute AA^T and find its eigenvalues.
 - (ii) Find orthonormal eigenvectors $\vec{u}_i$ (left singular vectors, columns of U). Feel free to use the associated Jupyter notebook to perform Gram-Schmidt for this part, if needed.
 - (iii) Find the singular values, $\sigma_i = \sqrt{\lambda_i}$.
 - (iv) Use $\vec{u}_i$ to find orthonormal $\vec{v}_i$ (for nonzero σ). Feel free to use the associated Jupyter notebook to perform Gram-Schmidt for this part, if needed.

- (v) Use the previous parts to write the full SVD of A .

- (b) Find the rank of A , using the computed full SVD.

- (c) Find a basis for the range (or columnspace) of A .

- (d) Find a basis for the nullspace of A .

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